



Harvesting and Selling

Lesson Aim

Specify procedures for harvest and post-harvest treatment and formulate market strategies for vineyard products.

Harvesting

The harvest period is the most important of time of the year for the viticulturist. It is the occasion when the “fruits” of their labour produce the desired result i.e. a full harvest of quality grapes. However, it can also be the most anxious period for the grower. Problems during this important part of the agricultural process can have a disastrous impact on the price obtained for the crop and on the final wine product. The primary source of concern leading up to the harvest is weather. Deteriorating weather usually takes the form of imminent rains; however, the early onset of frosts can also pose a threat in cool climate production regions. The worst case scenario is a sudden hailstorm. Ripening, and hence softening grapes, can be damaged severely by an untimely hail fall. The grower can be forced to make difficult decisions at this stage. If the crop is harvested early to avoid damage from the weather, grape quality may be sacrificed. Sugars and other flavour-related compounds may not have the opportunity to fully develop in the case of an early harvest. The alternative choice is to permit full ripening but run the risk of weather damage before or during the harvest.

Harvesting is best completed on a dry day. Wet grapes are highly susceptible to post-harvest diseases e.g. mould/fungi. If grapes do get wet, shake them off to remove excess water. On warm days pick in the early morning. Try not to leave grapes sitting overnight before pressing.

Testing for Ripeness

The start of the ripening process takes place about eight weeks after flowering. The grapes begin to change colour – this point is called *véraison*. From *véraison* on, the sugar and acid levels can be measured to determine when to harvest the grapes.

The stage at which grapes are picked (hence their levels of acid, sugar etc.), will greatly influence what they are suitable to be used for. Table grapes need to be sweet, as do dried grapes. At different stages of ripening, the grape will produce different types of wines.



The Ripening Process

As grapes become ripe, the following happens:

- Level of acid in the fruit decreases.
- Level of sugar in the fruit increases.
- Flavour and aroma increase.

Flavour intensity is the result of both sugar content and acidity. Grapes picked too late may have elevated sugar levels but acid content will be lost. Good pulp acid levels will enhance flavour and add character and life to wine. Although acidity can be adjusted during the winemaking process, rising grape pH can be a prompt to begin harvest.

Acid and sugar content in a representative sample will give a very clear indication of ripeness. Representative samples should be taken randomly from bunches in the mid-row vines rather than the end plants.

Three tests are conducted to measure optimum levels of ripeness:

- Brix, which measures the percentage of sugar in the grape juice. The term Brix describes the quantity of dissolved solids in a liquid. The majority of dissolved solids in grapes are sugars. Brix ratings are, therefore, closely related to sugar content. Hydrometers and refractometers are used to take sugar readings. Optimum Brix ratings are in the range of 19 to 24, depending on the variety and the climate.
- Titratable acid (TA), which measures the acidity. TA is a measure of total acid in the grape juice expressed as the tartaric acid content. Optimum TA is the range of 0.60 to 0.85 (also depending on variety and climate).
- pH, which also measures the acidity of grape juice, but is not necessarily correlated with the amount of tartaric acid. Optimum pH levels are around 3.1 for white grapes and 3.4 for red grapes.

N.B. A low pH reading indicates high quantities of acid, e.g. 2.5 – 3.5. Higher pH readings, e.g. 3.5+ represent a decrease in acid and a more alkaline grape. Grape acid levels are often expressed as tartaric acid equivalents.

A Brix to TA ratio is used to indicate the balance between sweetness and acidity in wine grapes. An optimum Brix:TA ratio for many wines is between 30:1 and 35:1.

Some growers taste the grapes leading up to harvest. Developing a sense of flavour intensity can be important for making the final harvesting decision, however this is a skill that may take some time to acquire. Aroma is also a consideration as it indicates the present of volatile compounds that are important to flavour.

Influence of Weather

The harvest can be affected by the following weather conditions:

- In COLD seasons sugar levels can stay low and acid levels high, irrespective of how long the grapes are left on the vines. The longer they are left on, in the hope of the weather turning warm, the more likely it is that disease develops in the fruit. If grapes are too acid when harvested, they may be used for wine, but the quality of the wine will suffer.
- In WARM, HUMID, WET seasons, vegetative growth may continue extra late in the season, creating a very humid area around the fruit, increasing the chance of disease. If there is wet weather too close to harvest, the fruit can crack. When wet conditions persist, the grower may decide to harvest early, the result being a must with low sugar, high acid and weaker flavour.
- In a HOT, DRY season, sugar production is good, but acid may be too low.

Pathogen Life Cycles

Disease-causing organisms have life-cycles too. Fungi and bacteria do not grow year round in all conditions. Some grape pathogens require specific temperature ranges and climatic conditions to proliferate and cause disease. Growing and harvesting grapes outside of the peak growth cycle of pathogens will not only produce a crop with a lower rate of disease, but also reduce pesticide costs.

Harvest Techniques

The small grower may still pick by hand, whereas the large commercial grower may find it more beneficial to harvesting mechanically. The risk of damaging grapes is often increased when mechanical harvesters are used but this depends on the degree of ripeness of the grapes and also the variety grown. Mechanical harvesting is not recommended for wine styles made with over-ripe grapes. Hand picking can be more appropriate for table grapes or dried fruit. Both of these end-products rely heavily on physical appearance to attract premium market prices.

Hand picking is done as carefully as possible so berries are not damaged. Juice of broken berries will oxidise and spoil, particularly in warm temperatures. An abundance of juice from broken fruit on the grapes' surface can lead to very rapid and prolific post-harvest disease. Greatest oxidation occurs at 30 degrees C). For this reason, picking is best done on cool days early in the morning. Boxes of grapes should be placed in the shade as soon as picked.

Grapes used for wine should be crushed as soon as possible after picking to prevent contamination by unwanted micro-organisms and excessive oxidation. Oxidation is less of a problem with red grapes than with white grapes. Despite this, mechanical harvesting of the fine red wine grapes (e.g. Pinot Noir) is not recommended.

Selling Grapes

The major sales outlets for the viticulturist include wineries, processing factories (e.g. dried fruit, jam and juice processors), and domestic and international fresh produce wholesalers.

It is important for growers to establish consistent and reliable markets for their produce, especially in areas where competition is strong and prices are variable. Some growers put as much effort into establishing good marketing partnerships as they do in actually growing the grapes. During the slower months of winter and spring, time and energy is spent on exploring and establishing profitable marketing relationships, leaving the summer free for production obligations.

Following the development of a marketing connection, growers need to be very diligent regarding ensuring the supply of a quality product. The first rule is to ensure they are producing the type of product required by their end-user. For example, if growing grapes for a winery, are they fulfilling the requirements for the specific wine style? With this issue in mind, the viticulturist actually makes marketing decisions when the vines are planted. Is it the correct type for current market demands? Will quality expectations be met i.e. are the soil and climatic conditions conducive to producing quality in the selected grape variety? Will the costs of required production inputs be covered by the price received for the grape types planted?

Guidelines for Selling Grapes to a Winery

- Establish a good communication with the winery, making sure that you clearly understand their needs and have a good awareness of their intake program.
- Be professional with your buyers, establishing a trustworthy business relationship.
- Receive input from the winery regarding what they thought of the grapes you supplied. A visit to the winery in the autumn can achieve this goal.
- Supply a document containing details of your vineyard. Ensure this is a professional-looking, well-presented document. (See 'Preparing a Vineyard Résumé')
- Follow-up communications with your customers. This acts to establish a good purchasing relationship. Utilise written communications or email links.

Selecting a grape purchaser that shares your level of professionalism is also important. You are more likely to establish a profitable business relationship with someone who displays a mutual interest in success, is professional in intent, and embraces a business approach that is flexible enough to adapt to market forces. The viability of the relationship will be greatly enhanced if you set good foundations. Foremost is your knowledge of the grapes that you are selling. You must be able to clearly, and concisely, articulate details of your produce and operation. Knowledge of both regional and global markets can be of great benefit.

Preparing a Vineyard Résumé

The “vineyard résumé” is an opportunity to display your passion for viticulture, your enthusiasm for your own vineyard and your knowledge of wine production and the grape industry. It is an important marketing tool so take care that the document is free of mistakes and demonstrates good knowledge of the industry and a dedication to produce good quality grapes. The résumé will assist buyers with their selections. For example, some wineries may prefer grapes grown with a particular aspect or in specific soils – a résumé targeted for these wineries should clearly state these factors.

Information to provide on a vineyard résumé could include:

- Grape varieties produced – the reasons they were chosen, as well as mention of significant cultural factors such as clones or rootstocks.
- Age and size of the vineyard.
- Crop details such as tonnes of fruit produced and a history of what has been grown.
- Production information such as soil type, trellis style, site climate, and management.
- Mention special qualities of the vineyard e.g. unique stock used, on-going research areas to improve characteristics.

Attractive, high quality photos and graphics will greatly enhance the appearance of your résumé.

Accurate yield estimations are important to the grower’s marketing strategies. Simply put, growers need to be able to supply the produce they have been contracted to provide. Likewise, unsold grapes reduce profitability. By its very nature grape production will be variable and yields will fluctuate. A good grower, however, will be able to identify influences as they develop and be able to make fairly accurate yield predictions. Communication with contracted buyers regarding the likelihood of meeting quotas is important. It is understood that bad yields do occur, but giving the buyer the opportunity to seek grapes elsewhere to fill the shortfall will be appreciated and will preserve the relationship.

Strategies For Selling Grapes

How do you sell grapes? Implementing successful sales of grapes can be achieved by considering the following:

1. Identify the Customer Base

The core customers will be wineries, wholesale fruit sellers or processors depending on the type of grapes that are grown.

2. Summarise the Production

- What varieties/clones do you have?
- Ages of vines
- A summary of site – soil type, altitude, hours of sunlight
- Work out the price you want and the payment schedule that you want from your customer. Remember that unproven grapes will command a lower price.

3. Research the Marketplace

Selling grapes requires an understanding of the big picture, not just the price. Is your particular market segment growing? Can you compete with grapes from other areas? If selling to a winery will they work with you on quality; i.e. do they want two tonnes per acre or seven?

The Salesperson

Every salesperson should know:

- Details of the grapes they are selling including varietal attributes for each variety grown, its competition, its negative points (and how to overcome these),
- Where and how to find the all the necessary documents relating to the sale.

- The prices to charge and terms of sale,
- Procedure for making a sale (including using cash register, filling out order forms, writing receipts etc.).
- Company policies.
- How the grapes will be packaged; trucking and delivery options.
- How to keep records in order.

A good salesperson should possess the following characteristics:

- A good appearance
- A pleasant personality
- Courtesy and tact
- Enjoy selling
- A basic understanding of human nature (practical not theoretical, e.g. the ability to read people's body language)

Key Rules Every Salesperson Should Follow

- Research your buyer and product first.
- Highlight the benefits of a particular grape variety rather than the features, e.g. demonstrate that the variety being grown can be used to produce a premium wine or larger, more succulent raisins.
- If the purchaser disagrees, keep calm and try to determine, very specifically, what problems the purchaser foresees, e.g. small production volumes, unreliable harvest time. Put the problem into perspective by demonstrating the compensations, e.g. 'Yes, the production volume is very small but a limited vintage of premium wine will command a higher price.'
- Always keep control of the conversation; don't let yourself get into a defensive position. This is done by asking questions when the customer starts to take the offensive.
- Handle berries carefully.
- Get the customer to sample the grapes.
- If you need to, use the phone or calculator to buy thinking time.
- Try to close the sale; ask for a contract at the appropriate time, when the customer seems to be in a state of mind where he/she is likely to buy.
- Remember that the buyer is always right - without him or her you are not going to remain in business.

Contracts

Grape contracts are the safest way to sell grapes, particularly in the case of wine grapes. Contracts should not be seen as representing mistrust; they are an indicator of professionalism. Some items that might be included in a contract are:

- Cultivar and clone; quantity to be delivered; price; Brix; TA; pH.
- Transportation responsibilities – who does it and in what type of container? When?
- Who is responsible for off-loading and weighing?
- Method and time of payment.
- Penalties if grapes are below minimum quality.
- Level of rot and other material allowed.

Always be prepared to discuss the terms of a contract but when doing so:

- Understand the current and future market conditions and developing consumer trends.
- Have a firm understanding of your priorities and goals; be able to communicate them.
- Be realistic about what you are capable of delivering and what advantages your vineyard really offers.

Both parties need to establish what is important to them. There must be give and take on both sides. Quality standards, price and contract length are all important.

Selling Online

Some online trading schemes are being explored. Growers list information such as varietal mix, spray diary and estimated production tonnage on a website. Buyers can then log on and bid for the grapes that they want and the fruit is sold to the highest bidder.

Brokers

Brokers can be used to make the process of buying and selling easier. Brokers will match the buyer and seller, because the broker understands the needs of each party. They also understand the dynamics of the market and can provide information to both parties to allow them to make informed decisions. Brokers can also arbitrate between seller and buyer and assist with smooth communication between them.

The functions of a broker include:

Provision of Marketing Advice and Performance Information

Brokers guide buyers and sellers to determine fair market values, based on consideration of demand; grape quality; quantities and payment terms.

Avoidance of Inappropriate Matches

The fundamental role of a broker is to find as many appropriate matches as possible between buyers and vineyards. The broker must be in constant contact with their clients be they buyers or sellers.

Follow up

The broker must make sure that where samples have been provided they arrive, are tasted and (hopefully) result in sales. Once a sales contract has been agreed, the broker will then follow-up the contract at its various stages, and ensures that any problems can be resolved quickly and amicably.

Avoidance of an Up-Front Fee

The broker normally only charges a fee once the grapes have been delivered and payment received by the seller.

Long Range Planning

The broker can advise and guide a grower in ways of developing a suitable customer base for the grapes, by considering such things as likely price decreases or the length of contract to enter into.

Marketing

Marketing is essential to successful selling. Grape production is expanding worldwide and it is increasingly necessary for the viticulturist to adopt an aggressive marketing policy for their fruit. To set yourself apart as a viticulturist, focus on quality in everything you do. Marketing cannot be regarded as an inconvenience and secondary to the production process. The purpose of marketing is get your berry production known and purchased by customers.



Grape Production - Where?

In 1981, world production was over 61 million tonnes of **Grapes**. Some of the largest producers of grapes at that time were:

Italy 20%

France 14%

USSR 11%

Spain 8%

USA 7%

Turkey 6%

In 1998, world production of **Wine** was:

Italy 20%

France 20%

Spain 11%

USA 8%

Argentina 5%

Germany 4%

South Africa 3.5%

Australia 3%

In 2003, world production of **Wine** was:

France 18%

Italy 17%

Spain 13.7%

USA 9%

Argentina 4.5%

China 4%

Australia 4%

Germany 3%

In 2011, world production of **Wine** was:

France 19%

Italy 16%

Spain 12.5%

United States 7.2%

Argentina 5.8%

China 4.8%

Australia 4.2%



Chile 3.9%

South Africa 3.5%

Germany 3.4%

Marketing and Economics

Most vineyards will take around four years to reach a stage of production that is commercially significant. Because of the high establishment and maintenance costs, coupled with the long-term nature of a vineyard, it is essential to develop a realistic marketing plan before embarking on commercial production of grapes irrespective of their end use.

Establishment and operating costs will vary significantly, even within the same region, because of the cost of elements such as land, labour and machinery. Grape cultivar/variety and cultural methods (such as training systems, pest management) will affect production costs, which must also be considered. Vineyard profitability will depend on both grape yield and price. The yield is dependent on cultural variables, such as weather, training systems, soil etc. The price is a function of the market AND the grower's ability to market the grapes.

It is important to allocate a budget to marketing and work within it. It is equally important to identify who is responsible for marketing activities. The financial allocation for promotion and advertising falls within the marketing budget. Ask the following questions:

How much money will you allocate towards marketing your grapes?

- A specific amount of money per box
- A percentage of the sales
- A fixed amount each year

Who is responsible for marketing your grapes?

The first decision of a marketing strategy is to identify the target market. You could ask the following questions:

- Who are your most desirable customers?
- Is this segment of the market profitable?
- Does it offer growth potential?
- Do you have a competitive advantage?
- Is this advantage sustainable?

The four P's traditional to all marketing are applicable to grapes: Product, Price, Promotion and Place.

Product

- Which type or varieties of grape will be grown?
- What will the berries be used for? (Juice, wine, table grapes etc.)
- What is the planned volume of production?
- Will the berries be hand-picked or mechanically harvested? (Usually hand-picked for table grapes and premium wine.)
- How will the grapes be packaged (table grapes)? Options include flat trays, bunch bags, and polystyrene (Styrofoam) boxes.

Price

- What is your pricing strategy? There are options such as high price/high quality or low price/high quality.
- How much does your production cost?
- What is the minimum price at which you can sell to break-even/make a profit?
- What are your competitors' prices? It is important that you understand your segment of the berry market. An undersupply of grapes will give you, the producer more control on prices.



Promotion

- Who do you want to target with your promotion activities?
- What media means will you use to target your market? For example, direct advertising in grocery stores, food magazines, brand identity via packaging.
- Will you allocate berry samples for promotion?
- Will you use discounts or special offers as promotional tools?

Place

- Is a distribution plan relevant?
- Will your customers come to you? (For example, pick-your-own table grapes; juicing co-operatives collect in bulk.)
- Can you sell only within your region? (For example, some American states restrict wineries to using only grapes grown in that state for wine production.)

Developing a Marketing Plan

A **Sample Marketing Plan** for **table grapes** might be developed as follows:

Target market

- Wholesale distributors; distributors selling to local supermarket/retail outlets or distributors for export market
- Direct growing contracts for large supermarket chains
- Pick-your-own; end consumers, health conscious, eating quality fresh fruit

Product

How many and which varieties?

One grape harvest per annum; grapes maturing at different times to extend the season – important for pick-your-own and for wholesale distributors.

Production Volume

Growing grapes for a co-operative/wholesale distributor may allow for smaller production volumes; volumes grown on contract for supermarkets will be tightly controlled.

Packaging Decisions

Some wholesale distributors will handle grapes packaged in Brand Identity boxes or bunch bags, such as “Joe’s Farm Table Grapes”; others will have their own distributor brand, such as “Just Grapes” (an Australian company which grows, packs and markets grapes); those operating as co-operatives are likely to have the co-operative branding on the packaging such as “Australian Table Grapes”, “Californian Seedless Table Grapes”.

Packaging materials vary from flat trays to open bunch bags or “zip-top” bags.

Price

Price structures will vary depending on how the grapes are sold, for example:

- Direct sales from grower to retail outlet – usually about two-thirds of retail price



- Bulk sales to wholesale distributor or co-operative – usually around one half of the retail price
- Farm-gate/pick-your-own sales – need to be set below the retail price to attract consumers

Promotion

- Direct sales to retailers rely on negotiating a contract before production starts and thus promotion is via past performance and quality.
- Grapes sold to wholesale distributors or co-operatives will usually be advertised and promoted by the wholesaler; the price paid to the grower will reflect this service. The advantage is that the wholesaler by virtue of larger capacity can target end consumers and the retailers.
- Farm-gate/pick-your-own will rely on local advertising in newspapers, on local TV and radio etc. Special event promotions can be effective.

Distribution or Place

- Direct sales to retailers: contract arrangements will specify whether the grower is responsible for packaging and delivery; the retailer collects in bulk and packages in-store etc.
- Growing for a wholesaler or co-operative: grapes are usually delivered by the grower.
- Pick-your-own/farm-gate: grapes are picked in situ – the vineyard is the venue.

Marketing Budget

- A fixed sum per tonne of grapes plus an additional sum for each of the other chosen means of advertising and promotion.
- Labour expended in marketing is not usually included in the marketing budget but added to the regular labour expenses.

Make Your Advertising Work

For advertising to be effective you need to develop strategies that are based on the “who”, “where”, “what” and “how” of your marketing strategy.

- Analyse your competitors: **Who** are your competitors? **How** do they reach their target market? What are the differences between you and your competitors?
- Know your target market: Have you identified **who and where** your customers are (i.e. your target market?) and **how** big your potential customer base is?
- Know how competitive you are: **What** position do you have in the market? Are you cheaper or more expensive than your competitors? Do you have a better or larger range of products?

Do you offer varieties, e.g. for premium wine, which are not offered by your competitors?

- Advertising budgets: **How** much money do you have to spend? **How** can you best reach your target market within your budget i.e. **What** can you afford – mail-outs, newspaper advertisements, magazine advertisements, radio, television, cable television, special offers?
- What do you want advertising to achieve? Will you concentrate on building a brand name for your grapes, or a quality standard under the vineyard name? In order for advertising to be effective, and to build your business, you need your target markets to act on the advertising i.e. pick up the phone or go to your premises. You therefore need to develop advertising messages that communicate clearly and frequently the wants and needs of your target audience. No amount of advertising creativity will get the message across if you have not identified these needs and desires.

Creative advertising can be generated in many ways and can be tailored to your budget and reflect your current advertising strategy:

- Build your advertising around a central idea, e.g. healthy eating.
- Offer “specials” this works well when you are trying to attract customers to your premises, e.g. for pick-your-own or farm-gate sales.
- Make it easy for your audience to become customers – include your location, phone number, internet site and email address and your business hours.

- If you have a brand logo – always use it in your advertising.
- Consider a 'Web Page' – web pages are useful even for businesses not offering online sales.
- Establish 'Web' links – make sure that wholesalers' and co-operatives websites provide a link to your vineyard; contact several search engines for their listing options.

Publicity Marketing

Publicity involves obtaining exposure for a product or service through the media, without paying for it. It may include such things as press releases which lead to radio interviews or articles in newspapers; eg. if you sign up to a new contract with a winery, it will benefit both the winery and the vineyard if the public knows about it.

Publicity is based on the following principles:

High Credibility

News or features seem more credible than an advertisement. If an article is written about a vineyard, this is a more credible way for someone to learn about that vineyard than to learn of it through an advertisement.

Catching People Off Guard

People, who might avoid advertisements or other sales approaches, will often read an article or listen to a news item about a new wine or a new vineyard.

Dramatising Products

Special, extraordinary events can be attached to product launches as a way of dramatizing the product. For example, offering free tasting days for new varieties of table grapes or new wines from a particular vineyard.

Market Research

Successful marketing depends upon knowing the people/groups you are marketing to; what they want, how they are likely to react to your varieties of grapes.

When the market place is understood, you can then follow the steps below to achieve successful marketing:

1. Set realistic marketing goals.
2. Provide structures for reaching those goals.
3. Assess the results of marketing efforts and modify your approaches accordingly.

Market research involves all those activities which help management reach marketing decisions. Market research attempts to make unknown things known; and in most instances, largely succeeds.

Types of Market Research

Market Research

Determining needs and wants of prospective customers; assessing the potential of specific market areas, studying the competition etc.

Product Research

Looking at grape varieties not currently being grown; testing prices to determine if they are too high or low etc.

Promotions Research

Checking effectiveness of displays or advertising.

Sales Research

Evaluating sales techniques of staff; looking at cost of selling; analysis of sales territory/distribution, etc.

Company Research

Determining industry trends; e.g. is wine production increasing or slowing? Determining the vineyard's "image" in the market place; studying employee morale or location of facilities.

Marketing Legalities

Being an alcoholic beverage, there are unique (and sometimes strict) legal controls upon the sale of wine in many countries. Many countries control not only the import, but also the production and export of wine. These controls can vary, from severe restrictions upon trade in wine, to taxes or tariffs designed to control their import or export. Similarly, many countries have strict quarantine controls on the import and export of fresh grapes.

Marketing Organisations

Organisations exist in many countries, for the expressed purpose of overseeing, controlling, or assisting in the marketing of grapes or grape products (including wine). Anyone working in the viticulture industry needs to familiarise themselves with relevant bodies in their country.

Where to Find More Information

The best way of finding out more about marketing relevant to your country is through an internet search. If you do not have the internet, try using either an internet café, or a local library. These exist all over the world today, and there are usually people at such places who can show you how to use the net.

Many people have trouble finding the right information on the internet because they don't know how to undertake a proper search

Two of the most common mistakes are:

- Searching on only one search engine.

(You need to realise that not all search engines are geared to search for all types of information. If you cannot find information fast, try using a different search engine.)

- Searching using only one word or phrase.

(A search for Viticulture in one country might bring far fewer results than a search for Grapes, or for Wine. In another country, Viticulture may turn up better results. Even subtle differences like the singular or plural of a word can turn up quite different results.)

